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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Empathy Coders
Selected Domain	Domain 2 - Orbyte Smart SME Challenge
Selected SDG Track (1–17)	8
Project / Solution Name	SmartOrder Hub
Name (Leader)	YANG JIAHUI
Email & Phone (Leader)	jacky666yang@gmail.com / +60 1123591658
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small businesses receive orders through a messy mix of WhatsApp messages, e-mail and spreadsheets. A coffee supplier may take one order on WhatsApp, another by email, then re-type both into a spreadsheet. The owner checks stock, guesses supplier lead time and calculates margin before promising delivery. This repetitive admin happens every evening and causes missed messages, wrong quantities, late deliveries and low-margin orders being accepted . POS and ERP systems are often too expensive or complex for a five-person business, while generic chatbots can read a message but cannot check inventory, calculate profit or route a human approval. The burden falls hardest on owners and staff with limited digital confidence, especially in low-margin retail, F&B and wholesale. Economically, this means lost sales, wasted stock and weaker productivity; socially, it increases working hours and stress. Domain 2 asks us to remove one recurring operational burden while keeping people in control.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SmartOrder Hub turns email, spreadsheet and simulated channel messages into one clean order table for Malaysian micro and small businesses. An AI extractor (DeepSeek V4.1 Flash) reads unstructured messages and fills customer, item, quantity, delivery and industry-specific fields; a deterministic rule parser is the fallback. Stock, supplier lead time, margin and a five-factor risk score are calculated by explicit rules, not AI. The owner approves, rejects or asks for information, while staff can review; every import, analysis, approval and status change is written to an activity log. It includes a read-only IMAP connector. Built with React, FastAPI and a JSON store, the prototype applies the SDC workshops' GenAI API, database and cloud-hosting skills and is deployed on Replit. WhatsApp and WeChat are not yet connected because their APIs require business accounts; they are future adapters. All data is synthetic, and Orbyte is a mock adapter.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript 7, Vite 8 and Tailwind CSS 4, with PapaParse for spreadsheet import. Backend: Python 3.11, FastAPI, Pydantic v2 and Uvicorn, with httpx, openpyxl and pytest. Data: a file-locked JSON document store with atomic writes and synthetic seed data; no external database server. Integrations: a real read-only IMAP connector, an optional DeepSeek V4.1 Flash chat-completions API, a deterministic rule-parser fallback and a mock OrbyteAdapter. WhatsApp and WeChat are not connected yet; their APIs require business accounts, so channel messages in the demo are simulated. Architecture: React SPA → FastAPI → extraction layer → rules and risk engine → JSON store. The prototype is deployed on the official SDC-provided Replit platform (Python 3.11 / Node 22); GitHub holds source and Replit Secrets holds credentials. Official SDC sponsor-provided cloud platform: Replit, used for build, hosting and deployment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking in five stages. Empathise: we studied the everyday pain points SMEs face when handling orders across WhatsApp, email and spreadsheets. Define: the recurring burden is not reading messages but re-keying, checking stock and deciding under time pressure. Ideate: we chose one clean order table with a single approval action instead of a complex ERP. Prototype: we built a React/FastAPI MVP with AI extraction, deterministic rules, risk scoring, role-based views and an activity log. Test: we ran a range of synthetic messages across six channels, checked edge cases such as missing fields and low margin, and refined the workflow after mentor feedback. Technical viability comes from a rules-only fallback, strict validation and automated tests; usability from three languages, clear risk explanations and owner-only approval. Deployment uses one FastAPI service on Replit, with GitHub for source and Replit Secrets for credentials. With the right business accounts and access, the system can connect to real WhatsApp/WeChat APIs in future.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.2:** "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." **Target 8.3:** "Promote development-oriented policies that support productive activities, decent job creation ,entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Baseline:** A representative five-person SME processing 20 orders/week spends about **3.7 hours/week** on manual order entry — nearly half a working day — based on a median of **11 minutes per order** (Esker, 2025). Source: Esker (2025). *Order Management KPI Benchmarking.* https://www.esker.com/blog/customer-service/how-order-management-benchmarks-can-guide-smarter-business-decisions/
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**KPI:** Median **minutes from customer message received to owner-approved order**, calculated automatically from activity-log timestamps.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final prototype presentation, reduce the median time from customer message received to owner-approved order by **70%** versus the manual baseline. Based on **11 minutes per order**, this means: Five-person SME, **20 orders/week**: **3.7 → ≤1.1 hours/week** Restaurant, **100 orders/week**: **18.3 → ≤5.5 hours/week**

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19; React DOM 19; Vite 8; @vitejs/plugin-react 6; Tailwind CSS 4; lucide-react 0.546; PapaParse 5.7	Code	MIT / ISC
FastAPI; Pydantic; Uvicorn; httpx; python-dotenv; openpyxl; pytest; google-genai (optional)	Code	MIT / BSD-3-Clause / Apache-2.0
DeepSeek V4.1 Flash; Google Gemini; OpenAI ChatGPT-6; Anthropic Claude Code	Model / AI tool	Proprietary / provider terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10/2026